

ABELEJE OLANIYI GEORGE

Software Engineer - Real-time AI & Voice Systems

olaniyigeorge77@gmail.com • [GitHub](#) • [Portfolio](#) • Lagos, Nigeria (Willing to relocate to SF)

SUMMARY

Backend engineer specializing in real-time AI voice systems and production LLM orchestration. Building Truefit.ai from the ground up, a dual-channel WebRTC/WebSocket voice interview platform with a custom async audio pipeline and a Gemini-powered tool-calling agent structured as a domain-isolated state machine. I write async Python naturally, reason about buffering and latency tradeoffs at the system level, and debug distributed failures end-to-end. Driven by the conviction that voice is the next frontier of human-AI interaction and that the engineering to get there is hard in exactly the right ways.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript

Backend: FastAPI, Django (DRF), Node.js, RESTful API Design, Webhooks

Real-Time & Async: WebRTC (aiortc), WebSocket, asyncio, event-driven architectures, message queues (Redis, BullMQ)

AI / LLM: Gemini Live API, OpenAI voice agents, tool-calling/function calling, prompt engineering, context management, LLM agent orchestration, RAG pipelines

Observability: Sentry, Datadog, structured logging, load testing, performance profiling

Cloud & DevOps: AWS (EC2, S3, RDS, SES, CloudFront), Docker, CI/CD, Render, Vercel

Databases: PostgreSQL, MongoDB, Redis, SQLite

Frontend: React, Next.js, Tailwind CSS

PROFESSIONAL EXPERIENCE

Fullstack Engineer — Winnov8

2025 – Present

- Designed and optimized a real-time notification and messaging system, improving throughput by ~58% under live production load.
- Implemented an intelligent recommendation engine, translating product requirements into reliable, testable backend services.
- Participated actively in architecture reviews, code reviews, and performance tuning for a growing production workload.

Fullstack Developer — Revela Africa

2025

- Built end-to-end event management features: ticketing, RSVP flows, secure payments, and user authentication.
- Integrated Google APIs and payment gateways with webhooks for automated onboarding and calendar syncing.
- Deployed Redis caching to stabilize service orchestration under peak traffic, significantly reducing response latency.

Backend Engineer — The ECO Platform

2024

- Audited and rewrote inefficient query paths, reducing API latency by over 91% and improving throughput to support thousands of concurrent users.

Fullstack Engineer — WispTalkAfrica

2023 – 2024

- Led full cloud migration to AWS (S3, RDS, EC2, CloudFront), significantly improving system uptime and resilience.
- Introduced monitoring practices and incremental performance improvements across infrastructure.

Freelance Software Developer — Independent

2020 – 2022

- Built automation tools, data scrapers, and web systems for clients across industries — foundation in backend engineering and API development.

PROJECTS

Truefit.ai: Real-time AI voice interview platform

Stack: Python, FastAPI, WebRTC (*aiortc*), Gemini Live API, *asyncio*, PostgreSQL, Redis, Next.js, TypeScript

- Architected a dual-channel real-time system: WebSocket for session signaling and WebRTC (SRTP/UDP) for all media — deliberately chosen over WebSocket audio streaming to eliminate head-of-line blocking, enable native echo cancellation, and ensure jitter-resilient delivery.
- Built a custom async audio pipeline in Python; diagnosed and fixed a compound production bug — a greedy queue drain collapsing 40 audio chunks into one `recv()` call, compounded by a pts clock reset corrupting frame pacing — resolving an infinite greeting loop that made the system unshippable.
- Structured Gemini Live as a tool-calling interview state machine with registered tools and domain-driven architecture using abstract ports, giving the agent zero dependency on infrastructure so providers are swappable without agent changes.
- Implemented a custom AudioBridge with PCM resampling and *asyncio* concurrency for multi-party audio session management.

CoopWise: AI-powered cooperative savings platform

Stack: FastAPI, PostgreSQL, Redis, Next.js, Gemini 1.5 Flash, Smart Contracts (Flow)

- Architected scalable FastAPI backend handling contribution cycles, payouts, and real-time financial state management.
- Integrated AI agents and RAG pipelines for personalized financial insights; implemented blockchain automation via Flow smart contracts.
- Earned hackathon recognition for system design, scalability, and real-world impact.

EDUCATION

B.Sc Computer Science — National Open University of Nigeria